

Pooja Kiran

Security Engineer | AI & Agent Security | Application Security | Cloud IAM Security | Model Supply-Chain Security
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EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 - Present

Self-Directed Research

Tempe, AZ, USA

- **Conducted security design reviews** across self-directed AI and cloud security systems, translating threat models into enforceable requirements, CI gates, and documented repository release criteria.
- Expanded the MCP Gateway validation track from **622 to 723 passing tests (+16.2%)** while statement coverage rose from **78.41% to 81.91% (+3.5 percentage points)** as measured Python statements grew from **4,679 to 5,644 (+20.6%)**.

Business & Compliance Lead, AEROSSEC

Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ, USA

- Led business/compliance analysis for a proposed airline PSS protection layer and built a **\$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

PROJECTS

Dates show original development periods; verification metrics below refer to current 2026 public-repository snapshots.

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security*

Oct. 2025 - Sep. 2026

- Secured the agent-to-tool execution boundary by engineering a default-deny MCP/JSON-RPC gateway with capability authorization, PII/exfiltration checks, anti-SSRF controls, rate limiting, and fail-closed enforcement before downstream execution.
- Expanded runtime detection with **55 prompt-injection patterns** and tamper-evident audit logging, producing reviewable security telemetry for policy decisions and incident analysis.
- Verified 2026 CI snapshot: **723 passing tests at 81.91% statement coverage**, plus **9 Elastic rules** and **21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Apr. 2025 - Sep. 2025

- Built a static IAM policy analyzer with **25 deterministic rules** covering wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy weaknesses, audit tampering, and permission boundaries.
- Made IAM findings reviewable in CI by emitting text, JSON, and SARIF 2.1.0 results into GitHub Code Scanning, turning policy weaknesses into actionable code-review evidence.
- Verified 2026 CI snapshot: **243 collected, 240 passed, 3 skipped**; the skipped checks require live AWS credentials.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras*

Nov. 2024 - Mar. 2025

- Built a non-executing model supply-chain scanner that inspects unsafe serialization and provenance indicators without executing untrusted repository code in the default inspection path.
- Combined pickle-opcode, AST/taint, dependency/provenance, obfuscation, and binary-format checks to surface risky behavior across SafeTensors, GGUF, ONNX, Keras, and pickle-derived artifacts.
- Verified 2026 CI snapshot: **241 passed, 1 skipped at 75.67% statement coverage**; detected **33/33 committed adversarial fixtures**.

EDUCATION

Arizona State University

Tempe, AZ, USA

M.S., Information Technology (Security), GPA: 3.87/4.00

Aug. 2024 - May 2026

- **Academic Participation:** Student participant, ASU, 2025; Generative AI learning assistant for personalized education.

M. S. Ramaiah University of Applied Sciences

Bengaluru, KA, India

B.Tech., Computer Science & Engineering, CGPA: 8.44/10

Aug. 2019 - Aug. 2023

TECHNICAL SKILLS

AI & Agent Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security

Identity & Application Security: Cloud IAM Security, AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Capability Authorization, Threat Modeling, API Security

Security Engineering & DevSecOps: Python, FastAPI, pytest, GitHub Actions, CI/CD Security, SARIF 2.1.0, GitHub Code Scanning, Docker

Detection & Observability: Elastic Security, ECS, SIEM, Prometheus, Tamper-Evident Audit Logging, MITRE ATLAS

CERTIFICATIONS

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)